



Awesense Energy Transition Platform Use Cases



- demoable implementation



- demoable & open sourced implementation

Jun 10, 2025



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UC01	Coincident Peak and Non-Coincident Peak Analysis		
UC02	Power Factor Analysis		
UC03	Phase Balance Analysis		
UC04	Transformer Capacity Analysis for EV Charging		
UC05	Reversed Power Flow and PV Capacity Analysis		
UC06	GIS Grid Connectivity Validation and Correction		
UC07	Asset Management - Cables and Lines		
UC08	Asset Management - Substations		
UC09	EV Charging and Use of Reserved Capacity Analysis		
UC10	Connected Loads [kW] vs. Actual Phase Balance Analysis		
UC11	Analysis of Unserved Energy Due to Outage Analysis		
UC12	Grid Temperature Monitoring		
UC13	Master Circuit Breaker Value vs. Measured Values Analysis		
UC14	Onsite Generation and Ghost Load Analysis		
UC15	Utility Data Challenge Hub		
UC16	Registered PV Generation Analysis		
UC17	Actual vs. Contracted Grid Capacity Analysis		
UC18	Grid Alerts and Warnings		
UC19	IoT Health Index and Overview		
UC20	Unexpected Generation Identification		
UC21	Geospatial Analysis of Grid Alerts and Warnings		
UC22	Topological Grid Segmentation		



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UC23	Analysis of Grid Segments and Elements			●
UC24	Analysis of Planned Outage for Assets Upgrades	🔓	●	
UC25	Analysis for Maximizing Value of IoT and Meter Devices			●
UC26	Transformer Loading Analysis	●		
UC27	IoT Sensor Billing	●		●
UC28	Energy Balancing for Revenue Protection			●
UC29	Non-Technical Loss Investigations Management			●
UC30	Suspicious Pattern Identification & Analysis			●
UC31	Grid and Time Series Visualization			●
UC32	Power Outage Visualization			●
UC33	MV Grid Optimization Analysis			
UC34	EV and Appliance Disaggregation			
UC35	Standard Consumption Patterns Creation			
UC36	Phase Identification Analysis			
UC37	Detection of Meter Incorrect Wiring Connection			
UC38	Transformer Load Increase Assessment Tool			
UC39	IPT and ICT Verification for Proper Billing			●
UC40	Phase Imbalance Correction Suggester	●		
UC41	EV Load Growth Forecasting	🔓		
UC42	Meter to Transformer Association Analysis			
UC43	Load Forecasting			
UC44	Voltage Power Quality Monitoring			●



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UC45 Feeder Voltage Analysis and Visualisation

UC46 Feeder Capacity and Lateral Capacity Analysis

UC47 Dynamic Line Rating Analysis

UC48 AMI-Smart Meter Flag Analysis for Theft Detection



UC49 Time Of Use Rate Analysis

UC50 Work Orders Management



UC51 Wildfire Risk Assessment

UC52 Hosting Capacity Analysis

UC53 Grid Map Publication Tool

[illegible]

[illegible]



Awesense Energy Transition Platform Use Cases - Focus Areas

	Consumer Rates & Programms	Situational Awareness and GridOps	Asset Management	Grid Planning	Data Quality Improvement	Billing	Outages	Distributed Energy Resources (DERs)	Multipurpose Data Analytics	Worker Safety	Electric Vehicles (EV)	Non-Wires Alternatives (NWA)	Revenue Protection
UC39 IPT and ICT Verification for Proper Billing				✓	✓							✓	
UC40 Phase Imbalance Correction Suggester	✓		✓								✓		
UC41 EV Load Growth Forecasting	✓		✓							✓			
UC42 Meter to Transformer Association Analysis		✓		✓				✓					
UC43 Load Forecasting	✓		✓										
UC44 Voltage Power Quality Monitoring		✓				✓							
UC45 Feeder Voltage Analysis and Visualisation		✓				✓							
UC46 Feeder Capacity and Lateral Capacity Analysis		✓	✓										
UC47 Dynamic Line Rating Analysis		✓									✓		
UC48 AMI-Smart Meter Flag Analysis for Theft Detection												✓	
UC49 Time Of Use Rate Analysis	✓				✓								
UC50 Work Orders Management		✓	✓			✓							
UC51 Wildfire Risk Assessment		✓	✓	✓		✓		✓					
UC52 Hosting Capacity Analysis	✓		✓	✓			✓			✓			
UC53 Grid Map Publication Tool	✓		✓			✓	✓			✓	✓		